

Solo E-Commerce Platform — Project Scope, Features & Requirements

Document Version: 1.0

Date: 17 March 2026

Author: Solo Engineering Team

Status: Final

1. Project Overview

1.1 Project Name

Solo E-Commerce — Premium Kitchenware & Home Goods Online Store

1.2 Project Description

A full-featured e-commerce web application serving the UAE market for premium kitchenware, tableware, glassware, and home goods. The platform supports end-to-end retail operations including product catalog management, shopping cart, online checkout with Stripe payments, order fulfillment, customer loyalty programs, and a content management system for marketing.

1.3 Target Market

- **Geography:** United Arab Emirates (UAE)
- **Currency:** AED (Arab Emirates Dirham)
- **Tax:** 5% VAT (standard UAE rate)
- **Language:** English (primary)
- **Product Focus:** Premium kitchenware brands (Eva Solo, Eva Trio, PWtbS, Eva)

1.4 Business Objectives

#	Objective	Success Metric
BO-1	Launch online retail presence	Website live with 800+ products
BO-2	Enable no-code content management	Admin can update homepage, banners, and landing pages without developer
BO-3	Automate order processing	End-to-end: browse → pay → order confirmed in < 5 minutes

#	Objective	Success Metric
BO-4	Build customer loyalty	Loyalty points program drives repeat purchases
BO-5	Reduce manual operations	Admin dashboard replaces spreadsheet-based inventory tracking
BO-6	Data-driven decisions	Reports dashboard provides revenue, product, and customer analytics

2. Project Scope

2.1 In Scope

Area	Description
Product Catalog	Browse, search, and filter 805+ products across categories, brands, and price ranges
User Accounts	Registration, login, profile management, address book, order history
Shopping Cart	Server-synced cart with quantity management, promo code application
Checkout & Payment	Multi-step checkout with Stripe payment integration (card payments)
Order Management	Order lifecycle from creation to delivery, with status tracking and invoices
Admin Dashboard	Product CRUD, order management, customer management, banner management
Content Management	Homepage sections, hero banners, landing pages, blog posts
Promotions	Promo codes (percentage, fixed, free shipping) with validation rules
Loyalty Program	Points earned per purchase, redeemable on future orders
Favorites/Wishlist	Save products for later viewing
Search	Text-based product search across name and description
Reports & Analytics	Revenue, orders, products, customers, VAT, and category reports
Media Management	Image upload, optimization, and CDN-ready serving
Navigation Management	Admin-configurable menu system
VAT Configuration	Admin-configurable VAT rate and calculation
Stripe Configuration	Admin-configurable payment gateway settings

Area	Description
Email Notifications	Order confirmation, password reset, email verification
Security	JWT authentication, RBAC, rate limiting, OWASP compliance

2.2 Out of Scope (Current Release)

Area	Reason
Mobile native apps (iOS/Android)	Web-first approach; Flutter supports future mobile build
Multi-language (Arabic)	Phase 2 consideration
Multi-currency	UAE market only (AED)
Marketplace (multi-vendor)	Single-vendor model
Live chat / customer support widget	External tool integration planned
SMS notifications	Email-only in current release
Subscription/recurring orders	Not required for current product types
Product reviews & ratings	Phase 2 feature
Advanced recommendation engine	Phase 2 (ML-based)
Warehouse management system	External WMS integration planned
Accounting software integration	Manual export via reports currently

3. Functional Requirements

3.1 Customer-Facing Requirements

FR-001: User Registration & Authentication

ID	Requirement	Priority
FR-001.1	Users can register with email and password	Must Have
FR-001.2	Users receive email verification after registration	Must Have
FR-001.3	Users can log in with email/password	Must Have
FR-001.4	Users can reset forgotten passwords via email	Must Have
FR-001.5	Users can change their password when logged in	Must Have

ID	Requirement	Priority
FR-001.6	Session persists across browser refreshes (JWT stored securely)	Must Have
FR-001.7	Tokens auto-refresh before expiration	Must Have
FR-001.8	Users can log out (revokes refresh token)	Must Have

FR-002: Product Browsing

ID	Requirement	Priority
FR-002.1	Homepage displays featured products, best sellers, and new arrivals	Must Have
FR-002.2	Users can browse products by category	Must Have
FR-002.3	Users can filter products by category, brand, and price range	Must Have
FR-002.4	Users can search products by name and description	Must Have
FR-002.5	Product listing shows image, name, brand, price	Must Have
FR-002.6	Products paginate at 20 items per page	Must Have
FR-002.7	Products can be sorted (price, name, newest)	Should Have
FR-002.8	Category landing pages display curated content	Should Have

FR-003: Product Detail

ID	Requirement	Priority
FR-003.1	Product page shows name, brand, images, price, description	Must Have
FR-003.2	Product page shows multiple images (gallery)	Must Have
FR-003.3	Product page shows specifications table	Should Have
FR-003.4	Product page shows dimensions and packaging info	Should Have
FR-003.5	Product page shows related products	Should Have
FR-003.6	Users can add product to cart from detail page	Must Have
FR-003.7	Users can add/remove product from favorites	Must Have
FR-003.8	Product page shows delivery and returns information	Should Have

FR-004: Shopping Cart

ID	Requirement	Priority
FR-004.1	Users can add products to cart with quantity	Must Have
FR-004.2	Cart persists server-side (survives logout/login)	Must Have
FR-004.3	Users can update item quantities in cart	Must Have
FR-004.4	Users can remove items from cart	Must Have
FR-004.5	Users can clear entire cart	Must Have
FR-004.6	Cart shows subtotal, VAT, and total	Must Have
FR-004.7	Users can apply promo codes in cart	Must Have
FR-004.8	Cart shows product images and names for each item	Must Have

FR-005: Checkout & Payment

ID	Requirement	Priority
FR-005.1	Multi-step checkout: address → payment → review → confirm	Must Have
FR-005.2	Users can select existing shipping address or add new	Must Have
FR-005.3	Users can select existing billing address or add new	Must Have
FR-005.4	Payment via Stripe (credit/debit card)	Must Have
FR-005.5	Order confirmation displayed on success	Must Have
FR-005.6	Order confirmation email sent	Must Have
FR-005.7	Promo code discount applied to order total	Must Have
FR-005.8	VAT calculated and displayed (5%)	Must Have
FR-005.9	Loyalty points awarded after order completion	Should Have

FR-006: User Account

ID	Requirement	Priority
FR-006.1	Users can view and edit their profile (name, phone)	Must Have
FR-006.2	Users can manage multiple addresses	Must Have
FR-006.3	Users can set a default shipping address	Must Have
FR-006.4	Users can view order history	Must Have
FR-006.5	Users can view individual order details	Must Have

ID	Requirement	Priority
FR-006.6	Users can download order invoices (PDF)	Should Have
FR-006.7	Users can view loyalty points balance	Should Have
FR-006.8	Users can save payment methods	Nice to Have

FR-007: Favorites / Wishlist

ID	Requirement	Priority
FR-007.1	Users can add/remove products from favorites	Must Have
FR-007.2	Favorites page shows all saved products	Must Have
FR-007.3	Favorite status shown on product cards and detail page	Must Have
FR-007.4	Toggle favorite from product card (heart icon)	Must Have

FR-008: Search

ID	Requirement	Priority
FR-008.1	Search bar accessible from header on all pages	Must Have
FR-008.2	Search queries product name and description (case-insensitive)	Must Have
FR-008.3	Search results show in product grid format	Must Have
FR-008.4	No results state shows helpful message	Must Have

3.2 Admin-Facing Requirements

FR-100: Admin Dashboard

ID	Requirement	Priority
FR-100.1	Dashboard shows total revenue (current month)	Must Have
FR-100.2	Dashboard shows total orders (current month)	Must Have
FR-100.3	Dashboard shows new customers (current month)	Must Have
FR-100.4	Dashboard shows average order value	Must Have
FR-100.5	Dashboard shows top selling products	Should Have
FR-100.6	Dashboard shows recent orders list	Must Have
FR-100.7	Dashboard shows order status distribution	Should Have

ID	Requirement	Priority
FR-100.8	Dashboard shows revenue by category	Should Have

FR-101: Product Management

ID	Requirement	Priority
FR-101.1	Admin can view all products with pagination	Must Have
FR-101.2	Admin can create new products with all fields	Must Have
FR-101.3	Admin can edit existing products	Must Have
FR-101.4	Admin can delete products	Must Have
FR-101.5	Admin can upload product images	Must Have
FR-101.6	Admin can set product flags (featured, best seller, new)	Must Have
FR-101.7	Admin can set product pricing (selling, original, cost)	Must Have
FR-101.8	Admin can assign products to categories and brands	Must Have

FR-102: Category Management

ID	Requirement	Priority
FR-102.1	Admin can view, create, edit, delete categories	Must Have
FR-102.2	Admin can reorder categories	Should Have
FR-102.3	Admin can manage subcategories	Should Have

FR-103: Brand Management

ID	Requirement	Priority
FR-103.1	Admin can view, create, edit, delete brands	Must Have

FR-104: Order Management

ID	Requirement	Priority
FR-104.1	Admin can view all orders with filtering by status	Must Have
FR-104.2	Admin can view order details (items, addresses, payment)	Must Have
FR-104.3	Admin can update order status with tracking info	Must Have
FR-104.4	Admin can generate and download invoice PDFs	Must Have

ID	Requirement	Priority
FR-104.5	Order status history is recorded (audit trail)	Must Have

FR-105: Customer Management

ID	Requirement	Priority
FR-105.1	Admin can view all customers with search/filter	Must Have
FR-105.2	Admin can view customer details and order history	Must Have
FR-105.3	Admin can create new customer accounts	Should Have
FR-105.4	Admin can edit customer profiles	Must Have
FR-105.5	Admin can soft-delete customers	Must Have
FR-105.6	Admin can adjust customer loyalty point balances	Should Have
FR-105.7	Admin can manage customer addresses	Should Have

FR-106: Banner Management

ID	Requirement	Priority
FR-106.1	Admin can create banners with title, subtitle, images, CTA	Must Have
FR-106.2	Admin can assign banners to placements (hero, top, mid, bottom)	Must Have
FR-106.3	Admin can set banner active dates (start/end)	Should Have
FR-106.4	Admin can enable/disable individual banners	Must Have
FR-106.5	Admin can reorder banners within a placement	Must Have

FR-107: Promo Code Management

ID	Requirement	Priority
FR-107.1	Admin can create promo codes (percentage, fixed, free shipping)	Must Have
FR-107.2	Admin can set minimum order amount for promo	Should Have
FR-107.3	Admin can set usage limits (total and per user)	Must Have
FR-107.4	Admin can set validity period (start/end dates)	Must Have
FR-107.5	Admin can activate/deactivate promos	Must Have
FR-107.6	Admin can view orders that used each promo code	Should Have

FR-108: Content Management (CMS)

ID	Requirement	Priority
FR-108.1	Admin can configure homepage sections and order	Must Have
FR-108.2	Admin can create/edit landing pages	Should Have
FR-108.3	Admin can configure category-specific landing pages	Should Have
FR-108.4	Admin can manage blog posts with categories and tags	Should Have
FR-108.5	Admin can configure navigation menus	Should Have
FR-108.6	Admin can manage curated product collections	Should Have

FR-109: Reports & Analytics

ID	Requirement	Priority
FR-109.1	Revenue report with date range filtering	Must Have
FR-109.2	Order report (count, status breakdown, trends)	Must Have
FR-109.3	Product report (top sellers, revenue by product)	Must Have
FR-109.4	Customer report (new vs returning, top spenders)	Should Have
FR-109.5	VAT report (taxable amount, VAT collected)	Must Have
FR-109.6	Category performance report	Should Have
FR-109.7	Promo code usage report	Should Have
FR-109.8	Stock/inventory status report	Should Have

FR-110: Settings Management

ID	Requirement	Priority
FR-110.1	Admin can configure VAT rate	Must Have
FR-110.2	Admin can configure Stripe API keys	Must Have
FR-110.3	Admin can enable/disable Stripe payments	Must Have

FR-111: Media Management

ID	Requirement	Priority
FR-111.1	Admin can upload images (JPEG, PNG, WebP, SVG)	Must Have

ID	Requirement	Priority
FR-111.2	Images auto-optimized on upload (WebP, resize)	Should Have
FR-111.3	Admin can delete uploaded media	Must Have
FR-111.4	Max file size: 5 MB	Must Have

4. Non-Functional Requirements

4.1 Performance

ID	Requirement	Target
NFR-001	API response time (p95)	< 500ms
NFR-002	Page load time (initial)	< 3 seconds
NFR-003	Concurrent users support	100+
NFR-004	Product listing pagination	20 items/page
NFR-005	Image load time	< 1 second (optimized + CDN)

4.2 Security

ID	Requirement	Standard
NFR-010	Password hashing	Argon2id
NFR-011	Transport encryption	TLS 1.2+
NFR-012	JWT token expiry	Access: 15min, Refresh: 7 days
NFR-013	Rate limiting	1000 req/min general, stricter on auth
NFR-014	Input validation	All inputs validated via DTOs
NFR-015	SQL injection prevention	Parameterized queries (Prisma)
NFR-016	XSS prevention	Helmet CSP headers, no raw HTML render
NFR-017	CORS	Whitelisted frontend origin only
NFR-018	Payment security	PCI-DSS via Stripe (no card data storage)

4.3 Reliability

ID	Requirement	Target
NFR-020	Uptime SLA	99.9%
NFR-021	Database backup frequency	Daily automated
NFR-022	Recovery point objective (RPO)	< 1 hour
NFR-023	Recovery time objective (RTO)	< 1 hour
NFR-024	Error handling	Graceful degradation, user-friendly messages

4.4 Usability

ID	Requirement	Target
NFR-030	Responsive design	Desktop, tablet, mobile viewports
NFR-031	Browser support	Chrome, Safari, Firefox, Edge (last 2 versions)
NFR-032	Accessibility	WCAG 2.1 AA target
NFR-033	Error messages	Clear, actionable user feedback

4.5 Maintainability

ID	Requirement	Target
NFR-040	Code structure	Modular (NestJS modules, Flutter providers)
NFR-041	API documentation	All endpoints documented
NFR-042	Database migrations	Version-controlled (Prisma Migrate)
NFR-043	Deployment	Automated CI/CD pipeline

5. Feature Implementation Matrix

Feature	Design	Backend	Frontend	Testing	Status
User Registration	✓	✓	✓	✓	Complete
User Login	✓	✓	✓	✓	Complete
Password Reset	✓	✓	✓	✓	Complete
Email Verification	✓	✓	✓	✖	Backend + Frontend Done
Product Listing	✓	✓	✓	✓	Complete

Feature	Design	Backend	Frontend	Testing	Status
Product Filtering	✓	✓	✓	✓	Complete
Product Detail	✓	✓	✓	✓	Complete
Product Search	✓	✓	✓	✓	Complete
Shopping Cart	✓	✓	✓	✓	Complete
Checkout Flow	✓	✓	✓	✓	Complete
Stripe Payments	✓	✓	✓	✓	Complete
Order Management	✓	✓	✓	✓	Complete
Invoice PDF	✓	✓	✓	⬜	API Complete
Favorites	✓	✓	✓	✓	Complete
Loyalty Program	✓	✓	✓	✓	Complete
Promo Codes	✓	✓	✓	✓	Complete
Admin Dashboard	✓	✓	✓	✓	Complete
Admin Products	✓	✓	✓	✓	Complete
Admin Categories	✓	✓	✓	✓	Complete
Admin Brands	✓	✓	✓	✓	Complete
Admin Orders	✓	✓	✓	✓	Complete
Admin Customers	✓	✓	✓	✓	Complete
Banners CMS	✓	✓	✓	✓	Complete
Landing Pages	✓	✓	✓	⬜	Frontend In Progress
Blog	✓	✓	⬜	⬜	Backend Complete
Navigation Menus	✓	✓	✓	⬜	Functional
Collections	✓	✓	⬜	⬜	Backend Complete
Reports	✓	✓	✓	⬜	Functional
Media Upload	✓	✓	✓	✓	Complete
VAT Config	✓	✓	✓	✓	Complete
Stripe Config	✓	✓	✓	✓	Complete

Legend: ✓ Complete | ⬜ Pending

6. User Stories Summary

6.1 Customer Stories

ID	Story	Priority
US-01	As a visitor, I want to browse products by category so I can find what I need	Must
US-02	As a visitor, I want to search for products so I can find specific items quickly	Must
US-03	As a visitor, I want to view product details with images and specs so I can make informed decisions	Must
US-04	As a customer, I want to register an account so I can save my preferences	Must
US-05	As a customer, I want to add items to cart and check out so I can purchase products	Must
US-06	As a customer, I want to pay securely with my credit card so my payment info is safe	Must
US-07	As a customer, I want to save my favorite products so I can buy them later	Must
US-08	As a customer, I want to view my order history so I can track my purchases	Must
US-09	As a customer, I want to manage my addresses so checkout is faster	Must
US-10	As a customer, I want to earn loyalty points so I get rewarded for purchasing	Should
US-11	As a customer, I want to use promo codes so I can get discounts	Should
US-12	As a customer, I want to download invoices so I have records of my purchases	Should

6.2 Admin Stories

ID	Story	Priority
US-20	As an admin, I want a dashboard showing key metrics so I can monitor business health	Must

ID	Story	Priority
US-21	As an admin, I want to manage products so I can keep the catalog current	Must
US-22	As an admin, I want to manage orders and update statuses so customers know their order progress	Must
US-23	As an admin, I want to manage customers so I can provide support	Must
US-24	As an admin, I want to create promo codes so I can run promotions	Should
US-25	As an admin, I want to manage homepage banners so I can promote products	Must
US-26	As an admin, I want to view reports so I can make data-driven decisions	Must
US-27	As an admin, I want to configure payment settings so I can manage Stripe	Must
US-28	As an admin, I want to upload product images efficiently so the catalog looks professional	Must

7. Data Requirements

7.1 Product Data

Field	Type	Required	Description
Name	String	Yes	Product display name (max 500 chars)
Slug	String	Yes	URL-friendly identifier
Description	Text	No	Full product description
Short Description	Text	No	Summary for listing cards
Brand	Reference	Yes	FK to Brand
Category	Reference	Yes	FK to Category
Subcategory	Reference	No	FK to Subcategory
Selling Price	Decimal	Yes	Current price in AED
Original Price	Decimal	No	Before-discount price
Cost Price	Decimal	No	Internal cost (admin only)

Field	Type	Required	Description
Currency	String	Yes	Default: AED
VAT Rate	Decimal	Yes	Default: 5%
Images	Array	Yes	1+ product images with display order
Specifications	Array	No	Key-value specification pairs
Dimensions	Object	No	Width, height, depth, unit
Is Active	Boolean	Yes	Show in store
Is Featured	Boolean	No	Show in featured section
Is Best Seller	Boolean	No	Show in best sellers
Is New	Boolean	No	Show in new arrivals
Designer	Reference	No	FK to Designer
Country	Reference	No	FK to Country of origin

7.2 Order Data

Field	Type	Required	Description
Order Number	String	Yes	Auto-generated: "SO-{timestamp}{random}"
Customer	Reference	Yes	FK to User
Status	Enum	Yes	PENDING → CONFIRMED → PROCESSING → SHIPPED → DELIVERED
Items	Array	Yes	Snapshot of ordered products
Subtotal	Decimal	Yes	Sum of item totals
Discount	Decimal	No	Applied promo discount
VAT Amount	Decimal	Yes	Calculated VAT
Shipping Amount	Decimal	No	Delivery charges
Total	Decimal	Yes	Final payable amount
Payment Intent	String	No	Stripe payment reference
Shipping Address	JSON	Yes	Address snapshot (immutable)
Billing Address	JSON	Yes	Address snapshot (immutable)

Field	Type	Required	Description
Promo Code	Reference	No	FK to PromoCode if applied
Loyalty Earned	Integer	No	Points awarded for this order
Loyalty Redeemed	Integer	No	Points used on this order

8. Constraints & Assumptions

8.1 Constraints

#	Constraint
C-1	Single-tenant: one Solo store per deployment
C-2	AED currency only (no multi-currency)
C-3	English language only (current release)
C-4	Maximum file upload size: 5 MB
C-5	Stripe is the only payment gateway supported
C-6	PostgreSQL 14+ required
C-7	Node.js 18+ required for backend
C-8	Flutter 3.38+ required for frontend builds

8.2 Assumptions

#	Assumption
A-1	Products are physical goods with basic shipping (no digital downloads)
A-2	Single warehouse fulfillment model
A-3	Admin users are trusted internal staff (no audit for admin actions beyond orders)
A-4	Email delivery is critical for password reset and order confirmation
A-5	Product catalog is manually maintained by admin (no automated supplier feeds)
A-6	Internet connectivity is required for all operations (no offline mode)

9. Acceptance Criteria (Key Flows)

9.1 Customer Registration

- **Given** a visitor on the signup page
- **When** they enter valid email, password, first name, last name
- **Then** an account is created, verification email is sent, and they are redirected to verification prompt

9.2 Product Purchase

- **Given** a logged-in customer with items in cart
- **When** they complete checkout with valid shipping address and Stripe payment
- **Then** an order is created, cart is cleared, loyalty points are awarded, and confirmation email is sent

9.3 Admin Order Management

- **Given** an admin viewing order with status "CONFIRMED"
- **When** they update status to "SHIPPED" with tracking number
- **Then** order status is updated, status history entry is created, and customer can view new status

9.4 Promo Code Application

- **Given** a customer with a cart subtotal of AED 500
- **When** they apply a valid promo code for 10% off (min order AED 200)
- **Then** AED 50 discount is applied, new total reflects reduction

10. Release Plan

Phase 1 — MVP (Current Release)

- Full product catalog with search and filtering
- User registration, login, account management
- Shopping cart + checkout + Stripe payments
- Admin dashboard + product/order/customer management
- Banner CMS + homepage configuration
- Promo codes + loyalty program (basic)
- Reports dashboard

Phase 2 — Enhancements (Planned)

- Product reviews and ratings
- Arabic language support (RTL)

- Advanced search (filters UI, faceted search)
- Customer email marketing integration
- Blog frontend display
- Curated collections frontend
- Stock management alerts
- Shipping integration (Aramex / Emirates Post)

Phase 3 — Scale (Future)

- Mobile native shell (Flutter → Android/iOS)
- AI-powered product recommendations
- Multi-warehouse support
- Accounting software integration
- Advanced analytics (conversion funnels, cohort analysis)
- A/B testing framework for CMS content

End of Project Scope, Features & Requirements Document